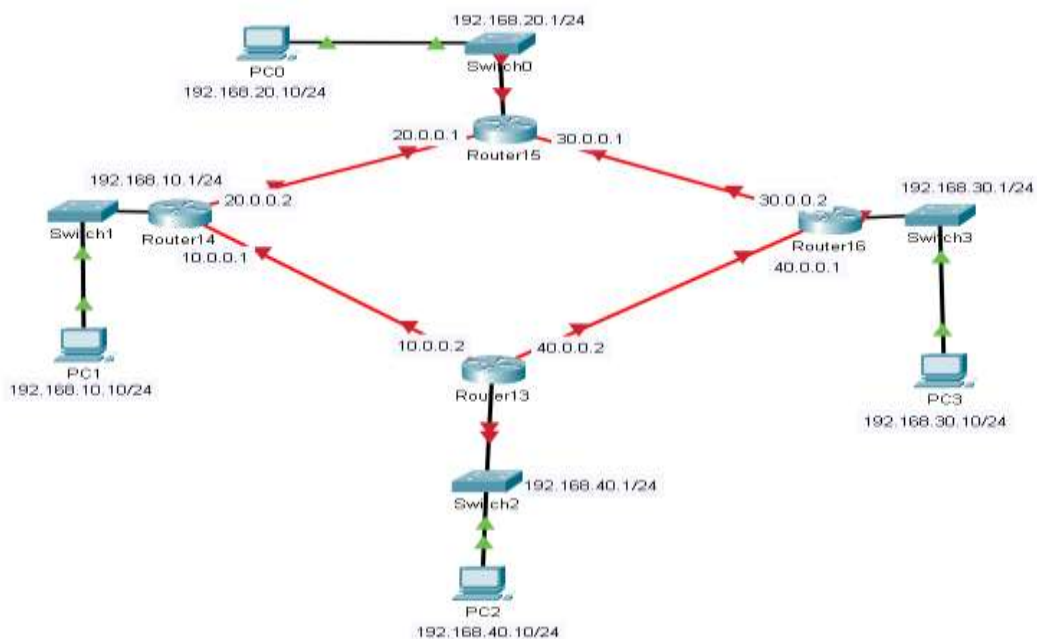


OSPF (Open Shortest Path First) is a link-state interior gateway protocol — an open standard (OSPFv2 for IPv4, RFC 2328; OSPFv3 for IPv6). Unlike distance-vector protocols that share their whole routing table with neighbors ("routing by rumor"), each OSPF router builds a full map of the topology and computes its own paths.

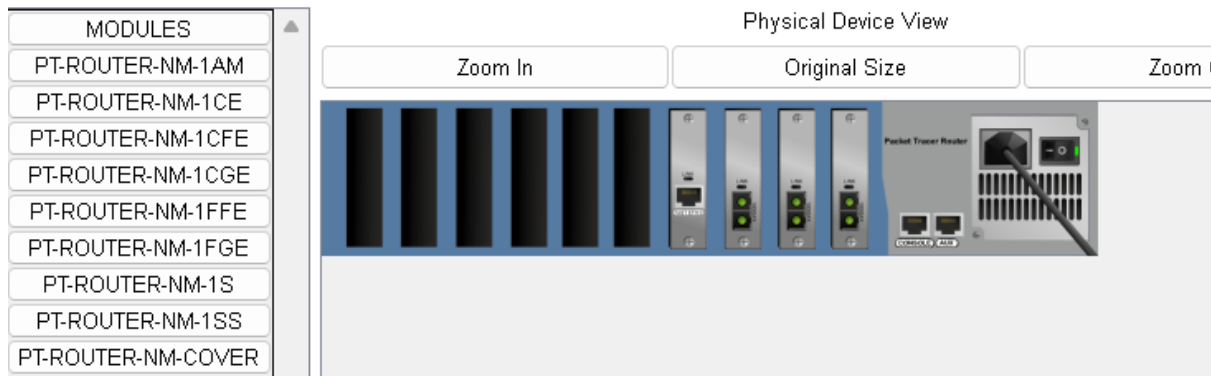
The mechanics, briefly:

- **Neighbors and LSAs.** Routers discover each other with Hello packets and then exchange Link-State Advertisements describing their links and costs. Every router in an area floods and stores these in a shared **link-state database (LSDB)** — so all routers in an area see an identical map.
- **SPF / Dijkstra.** Each router runs the Dijkstra shortest-path-first algorithm against that LSDB to compute the lowest-cost path to every destination. **Cost** is the metric, derived from bandwidth by default (lower cost = preferred), which is why OSPF makes smart bandwidth-aware decisions where a hop-count protocol like RIP wouldn't.
- **Areas.** OSPF scales through hierarchy. **Area 0** is the backbone; all other areas must connect to it, typically through an **ABR** (Area Border Router). Areas limit the scope of LSA flooding and SPF recalculation, which keeps large networks stable.
- **DR/BDR.** On multi-access segments (like your Switch4 segment), OSPF elects a Designated Router and Backup DR so routers don't have to form full adjacencies with each other in an n^2 mesh — they all adjacent to the DR instead. This is the specific behavior your topology showcases.

That's the topology



We going to add a non-empty router so we can add the interfaces cause we going to exceed the normal size of interfaces



I have added Cisco Gigabit Ethernet Network Module (part number PT-ROUTER-NM-1FGE) provides Gigabit Ethernet copper connectivity for access routers.

And The PT-ROUTER-NM-1CFE Module provides one Fast-Ethernet interface for use with copper media.

After applying the topology start on configuring the IP of each device from ip configuration

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::20A:41FF:FEE6:C433

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

And for routers

```

Router(config)#interface fa2/0
Router(config-if)#ip ad
Router(config-if)#ip address 192.168.10.1 255.255.255.0

Router(config-if)#ip address 20.0.0.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

```

After finishing all PCs start on configuring the routers and make sure you make the CIDR /30 to stop the waste of Subnet

10/20/30/40.0.0.0 aren't all private. Only 10.0.0.0/8 is RFC 1918 — 20, 30 and 40 are publicly routable ranges. Fine in a lab, but worth knowing so the habit doesn't follow you into a real design. A cleaner scheme would be 172.16.x.0/30 for each link.

Before router configuration we test the ping from 192.168.10.10 to 192.168.20.10

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

It fails

Let's do the router configuration

The Router Configuration:

!===== R15 (top)=====

interface FastEthernet3/0

ip address 192.168.20.1 255.255.255.0

no shutdown

interface GigabitEthernet0/0

ip address 20.0.0.1 255.255.255.252

no shutdown

interface GigabitEthernet1/0

```
ip address 30.0.0.1 255.255.255.252
```

```
no shutdown
```

```
!
```

```
router ospf 1
```

```
router-id 15.15.15.15
```

```
auto-cost reference-bandwidth 10000
```

```
passive-interface FastEthernet3/0
```

```
network 192.168.20.0 0.0.0.255 area 0
```

```
network 20.0.0.0 0.0.0.3 area 0
```

```
network 30.0.0.0 0.0.0.3 area 0
```

```
! ===== R14 (left) =====
```

```
interface FastEthernet2/0
```

```
ip address 192.168.10.1 255.255.255.0
```

```
no shutdown
```

```
interface GigabitEthernet1/0
```

```
ip address 20.0.0.2 255.255.255.252
```

```
no shutdown
```

```
interface GigabitEthernet0/0
```

```
ip address 10.0.0.1 255.255.255.252
```

```
no shutdown
```

```
!
```

```
router ospf 1
```

```
router-id 14.14.14.14
```

```
auto-cost reference-bandwidth 10000
```

```
passive-interface FastEthernet2/0
```

```
network 192.168.10.0 0.0.0.255 area 0
```

```
network 20.0.0.0 0.0.0.3 area 0
```

```
network 10.0.0.0 0.0.0.3 area 0
```

! ===== R16 (right) =====

interface FastEthernet3/0

ip address 192.168.30.1 255.255.255.0

no shutdown

interface GigabitEthernet1/0

ip address 30.0.0.2 255.255.255.252

no shutdown

interface GigabitEthernet0/0

ip address 40.0.0.1 255.255.255.252

no shutdown

!

router ospf 1

router-id 16.16.16.16

auto-cost reference-bandwidth 10000

passive-interface FastEthernet3/0

network 192.168.30.0 0.0.0.255 area 0

network 30.0.0.0 0.0.0.3 area 0

network 40.0.0.0 0.0.0.3 area 0

! ===== R13 (bottom) =====

interface FastEthernet3/0

ip address 192.168.40.1 255.255.255.0

no shutdown

interface GigabitEthernet0/0

ip address 10.0.0.2 255.255.255.252

no shutdown

interface GigabitEthernet1/0

ip address 40.0.0.2 255.255.255.252

no shutdown

!

router ospf 1

router-id 13.13.13.13

auto-cost reference-bandwidth 10000

passive-interface FastEthernet3/0

network 192.168.40.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

network 40.0.0.0 0.0.0.3 area 0

```
Passive Interface(s) :
  GigabitEthernet2/0
  FastEthernet3/0
```

```
Router#show ip ospf ne
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
14.14.14.14	1	FULL/BDR	00:00:30	20.0.0.2	GigabitEthernet0/0

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
15.15.15.15	1	FULL/DR	00:00:30	20.0.0.1	GigabitEthernet1/0

After configuring all above

The neighbours should see each other

Neighbor ID	Pri	State	Dead Time	Address	Interface
15.15.15.15	1	FULL/DR	00:00:39	30.0.0.1	GigabitEthernet1/0
13.13.13.13	1	FULL/BDR	00:00:30	40.0.0.2	GigabitEthernet0/0

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
16.16.16.16	1	FULL/BDR	00:00:33	30.0.0.2	GigabitEthernet1/0
14.14.14.14	1	FULL/BDR	00:00:30	20.0.0.2	GigabitEthernet0/0

After configuring all the ping should work as follows:

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```